Jun Xue

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EDUCATION

Anhui University **09/2021-Present**
Computer Science and Technology
M.Eng. in *Computer Science & Technology*

Courses: Machine Learning, Neural Networks, Combinatorial Mathematics, Modern signal processing technology, Pattern Recognition.

Anhui Science And Technology University **09/2016-06/2020**
School of Electronic Science and Technology
B.Eng. in *Electronic Science and Technology*
Courses: Analog Circuits, Digital Circuits, Circuit Analysis, C programming.

RESEARCH PROJECTS

Fake speech dectection **09/2021- Present**

- Finding discriminative acoustic features for anti-spoofing
- Modeling fake speech dectection tasks using deep neural networks
- Building a generalized and generic framework for fake speech detection.

Knowledge distillation

- Explore knowledge distillation methods for anti-spoofing
- Building high-performance, low-complexity fake speech detection systems.

RESEARCH

Current Work

- Fake speech detection
- Knowledge distillation
- Self-supervised learning

Papers：

[1] **Jun Xue**, et al. "Audio Deepfake Detection Based on a Combination of F0 Information and Real Plus Imaginary Spectrogram Features." Proceedings of the 1st International Workshop on Deepfake Detection for Audio Multimedia. 2022.
[2] **Jun Xue**, Fan C, Yi J, et al. Learning from yourself: A self-distillation method for fake speech detection[C]//ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). IEEE, 2023: 1-5.
[3] 王煜，**薛军**，范存航(*)，项铖，戎晨凯，张雪昆，李沼坤，吕钊(*)．基于自适应通道注意力的语音伪造检测方法．2022年第十七届全国人机语音通讯学术会议.
[4] Shunbo Dong, **Jun Xue**, et al. Multi-perspective Information Fusion Res2Net with Random Specmix for Fake Speech Detection. the IJCAI 2023 Workshop on Deepfake Audio Detection and Analysis (DADA 2023).

Under review：

[5] Cunhang Fan (*), **Jun Xue**, et al. Subband Fusion of Complex Spectrogram for Fake Speech Detection [J] Speech Communication. (Major revise)

[6] **Jun Xue**, et al. Beyond the Teacher: A Data-Guided Knowledge Distillation Method for Fake Speech Detection [J] IEEE Signal Processing Letters.

[7] Cunhang Fan (*), **Jun Xue**, et al. Spatial Reconstructed Local Attention Res2Net with F0 Subband for Fake Speech Detection [J] IEEE Transactions on Audio, Speech, and Language Processing.

Note: * indicates the corresponding author